Stage 1: Train the autoencoder

$$x \to \mathcal{E} \to (\mu, \sigma) \to z = \mu + \sigma \odot \eta \to \mathcal{D} \to \tilde{x}$$

Reconstruction loss + KL regularization; $\eta \sim \mathcal{N}(0, \boldsymbol{I})$

Stage 2: Freeze the autoencoder; train latent diffusion

$$x \to \mathcal{E} \to z_0 \to z_t \to \varepsilon_{\theta\left(z_t, t, \tau_{\theta(y)}\right)}$$

Add known Gaussian noise; optimize the noise-prediction error.

Generation: No input image is required

$$z_T \sim \mathcal{N}(0, \boldsymbol{I}) \quad \to \quad z_{T-1} \quad \to \quad \ldots \quad \to \quad z_0^{\text{gen}} \to \mathcal{D} \to x_{\text{gen}}$$

Repeat: predict noise, update latent, and add reverse noise if $t > 1$.

Condition $y \to \tau_{\theta(y)}$ supplies keys/values to the denoiser.